

LAB 5

CONFIGURATION OF STATIC ROUTES AND DEFAULT ROUTES

OBJECTIVES

- To understand the concepts of Static Routing and Default Routing for manual path selection in a network.
- To configure a network topology in Cisco Packet Tracer using Static and Default routes.
- To establish connectivity between different networks through Static and Default routing techniques.
- To visualize and analyze the flow of packets across networks to verify routing decisions.

THEORY

1. Static Routing

Static Routing is the process of manually configuring routing entries in a router's routing table. In this method, the network administrator specifies the exact path that data packets should follow to reach a destination network.

Operation: Static routes are configured using the command:

```
ip route [destination_network] [subnet_mask] [next_hop_address]
```

The router consults this entry to forward packets. If the defined link fails, the administrator must manually modify or update the route.

Application: Static routing is typically used in small networks with simple, stable topologies. It is also applied for security purposes, such as concealing parts of a network from external visibility.

2. Default Routing

Default Routing is a special form of static routing that serves as the “*gateway of last resort*.” In this method, the router forwards all packets to a designated next-hop router whenever the destination network is not explicitly listed in the routing table.

Operation: Default routes are configured using the command:

```
ip route 0.0.0.0 0.0.0.0 [next_hop_address]
```

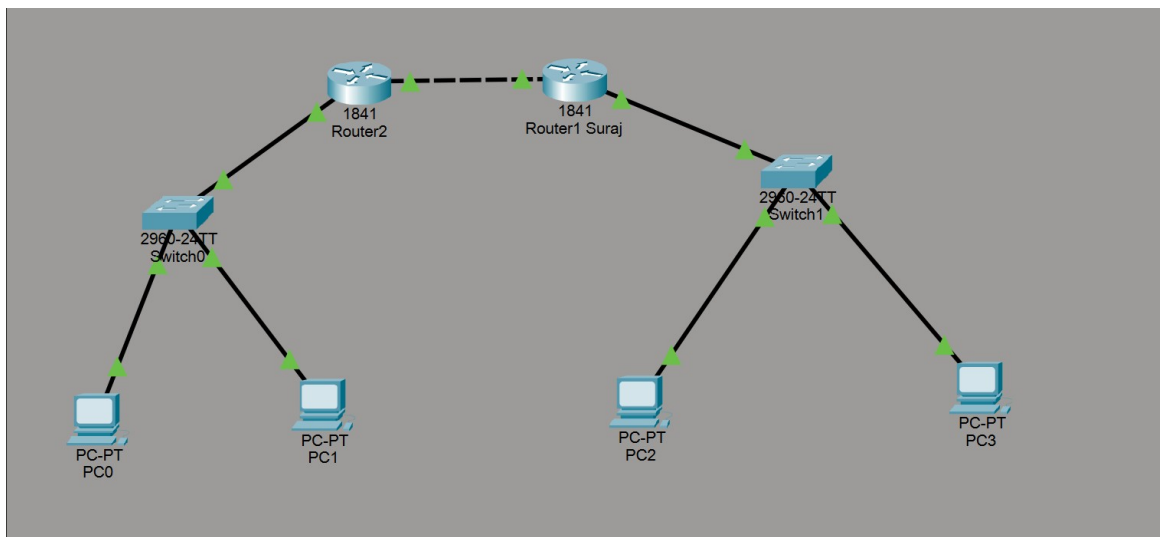
This instructs the router to send any traffic with an unknown destination to the specified neighbor.

Application: Default routing is mainly used in stub networks (networks with only one exit point) or for connecting a local network to the Internet via an ISP. It helps simplify routing by reducing the size of the routing table.

NETWORK TOPOLOGY

The network topology in the figure consists of two routers: Router2 and Router1 Suraj, connected via a serial link to enable communication between two networks.

- On the left side, Router2 is connected to Switch0, which links PC0 and PC1, forming the first LAN segment.
- On the right side, Router1 Suraj is connected to Switch1, which links PC2 and PC3, forming the second LAN segment.
- The serial connection between the two routers allows data packets to travel between the two LANs, enabling full inter-network communication.
- The connection between two router was done through cross-cabling.



CONFIGURATION TABLE:

Device	Interface	IPv4 Address	Subnet Mask	Default Gateway
Router2	GigabitEthernet0/0	10.0.0.1	255.0.0.0	N/A
Router2	GigabitEthernet0/1	192.168.1.1	255.255.255.0	N/A
Router1 Suraj	GigabitEthernet0/0	10.0.0.2	255.0.0.0	N/A
Router1 Suraj	GigabitEthernet0/1	192.168.2.1	255.255.255.0	N/A

PC0	FastEthernet0	192.168.1.2	255.255.255.0	192.168.1.1
PC1	FastEthernet0	192.168.1.3	255.255.255.0	192.168.1.1
PC2	FastEthernet0	192.168.2.2	255.255.255.0	192.168.2.1
PC3	FastEthernet0	192.168.2.3	255.255.255.0	192.168.2.1

Routing Configuration

Syntax :

```
ip route <destination_network> <subnet_mask> <next_hop_ip>
```

Device	Routing Type	Command / Route
Router2	Static Route	ip route 192.168.2.0 255.255.255.0 10.0.0.2
Router1 Suraj	Static Route	ip route 192.168.1.0 255.255.255.0 10.0.0.1
Router2	Default Route	ip route 0.0.0.0 0.0.0.0 10.0.0.2
Router1 Suraj	Default Route	ip route 0.0.0.0 0.0.0.0 10.0.0.1

RESULT

The network configuration was completed successfully. To test connectivity, a ping request was initiated from PC2 (192.168.2.2) in the second network to PC0 (192.168.1.2) in the first network. The packets were transmitted and received without error, confirming the correctness of the static route. A similar verification was carried out for the default route, which also functioned as expected.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Fig : Ping request (static routing)

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C:\>|
```

Fig : Ping request (dynamic routing)

DISCUSSION

In this experiment, static and default routing were configured to enable communication between two separate LANs connected through routers. The static route ensured that packets followed a manually defined path, while the default route acted as a gateway of last resort for unknown destinations. Testing with ping requests confirmed that both routing methods worked correctly, allowing seamless packet transfer between the networks. This demonstrates how routing choices affect connectivity and efficiency in different network scenarios.

CONCLUSION

The topology was successfully implemented using Cisco Packet Tracer, and both static and default routing were verified through connectivity tests. Static routing provided precise control over packet paths, while default routing simplified configuration in networks with a single exit point. Overall, the experiment validated the importance of routing techniques in ensuring reliable communication between networks.